Accounts Payable Value Framing Business Case

A Process Mining Analysis of P2P Deviations and Improvement Opportunities Using Celonis Methodology

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Why Accounts Payable? The Value Framework

SPEND:

- Incorrect payment rate
- Total discounts captured

→ Opportunities: Duplicate Invoices, Lost Cash Discounts

PRODUCTIVITY:

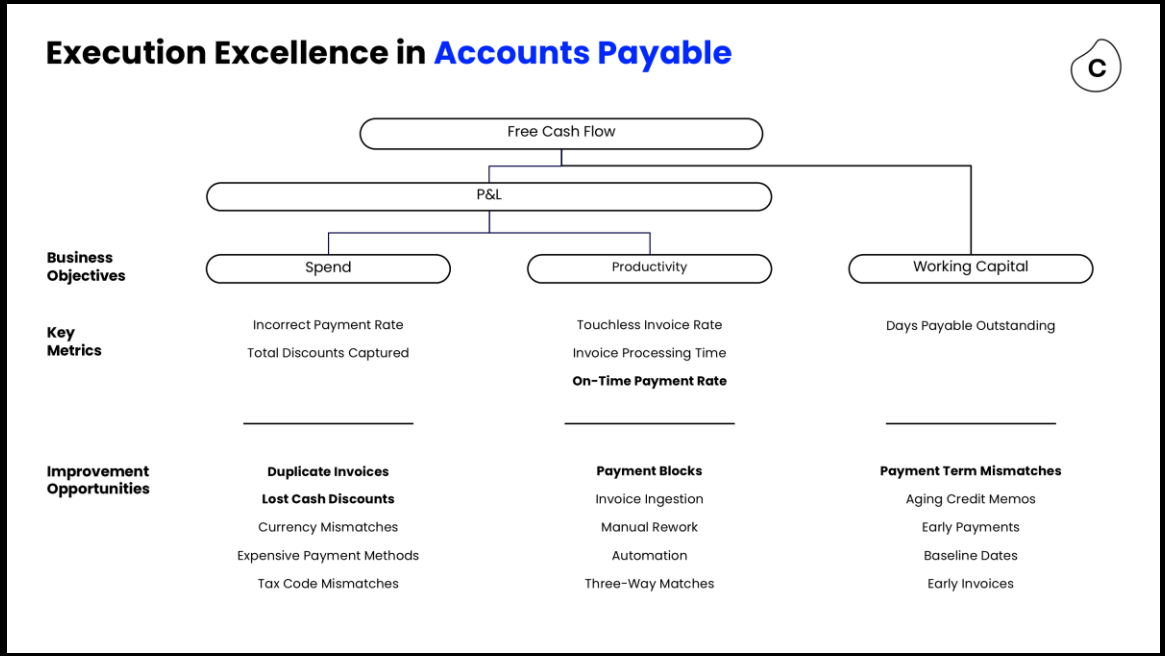
- Touchless invoice rate
- Invoice processing time
- On-time payment rate

→ Opportunities: Payment Blocks, Manual Rework

WORKING CAPITAL:

- Days Payable Outstanding

→ Opportunities: Payment Term Mismatches, Early Payments



"Source: Celonis AP Value Tree – Execution Excellence Framework

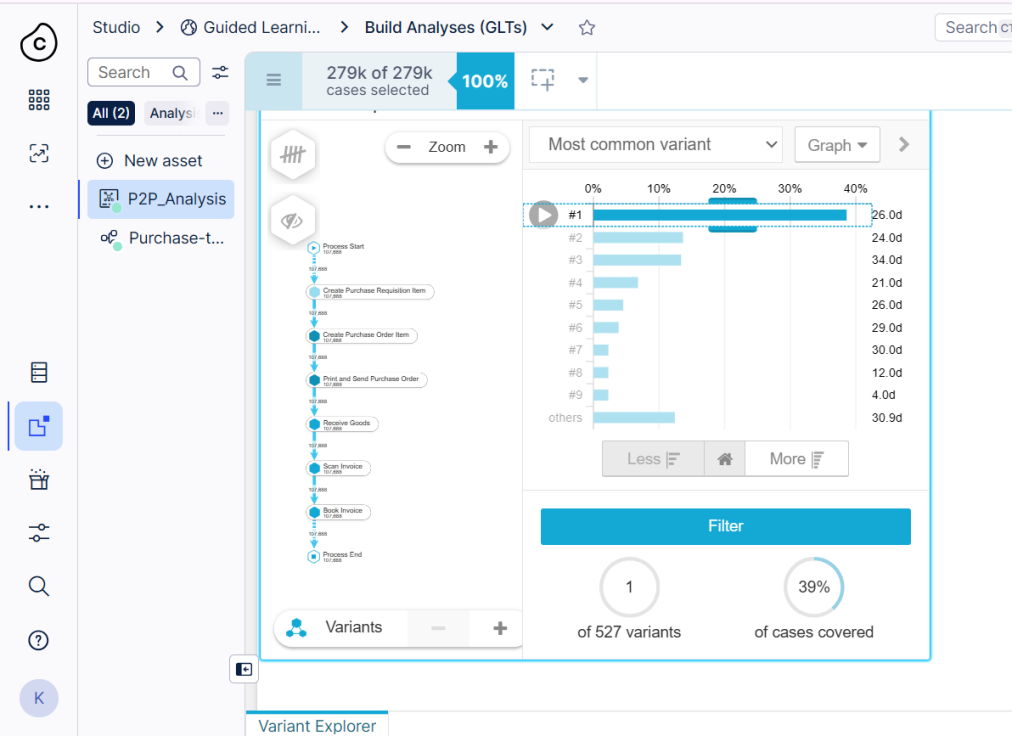
Both focus opportunities in this project target the Productivity pillar, directly impacting Free Cash Flow through P&L improvement

The Dataset: P2P Event Log — Key Statistics

812 Cases | 4,927 Events | 20 Vendors | 29.6 Days

Process Flow

Create PR → Create PO → Send PO → Goods receipt → Invoice receipt → Clear invoice



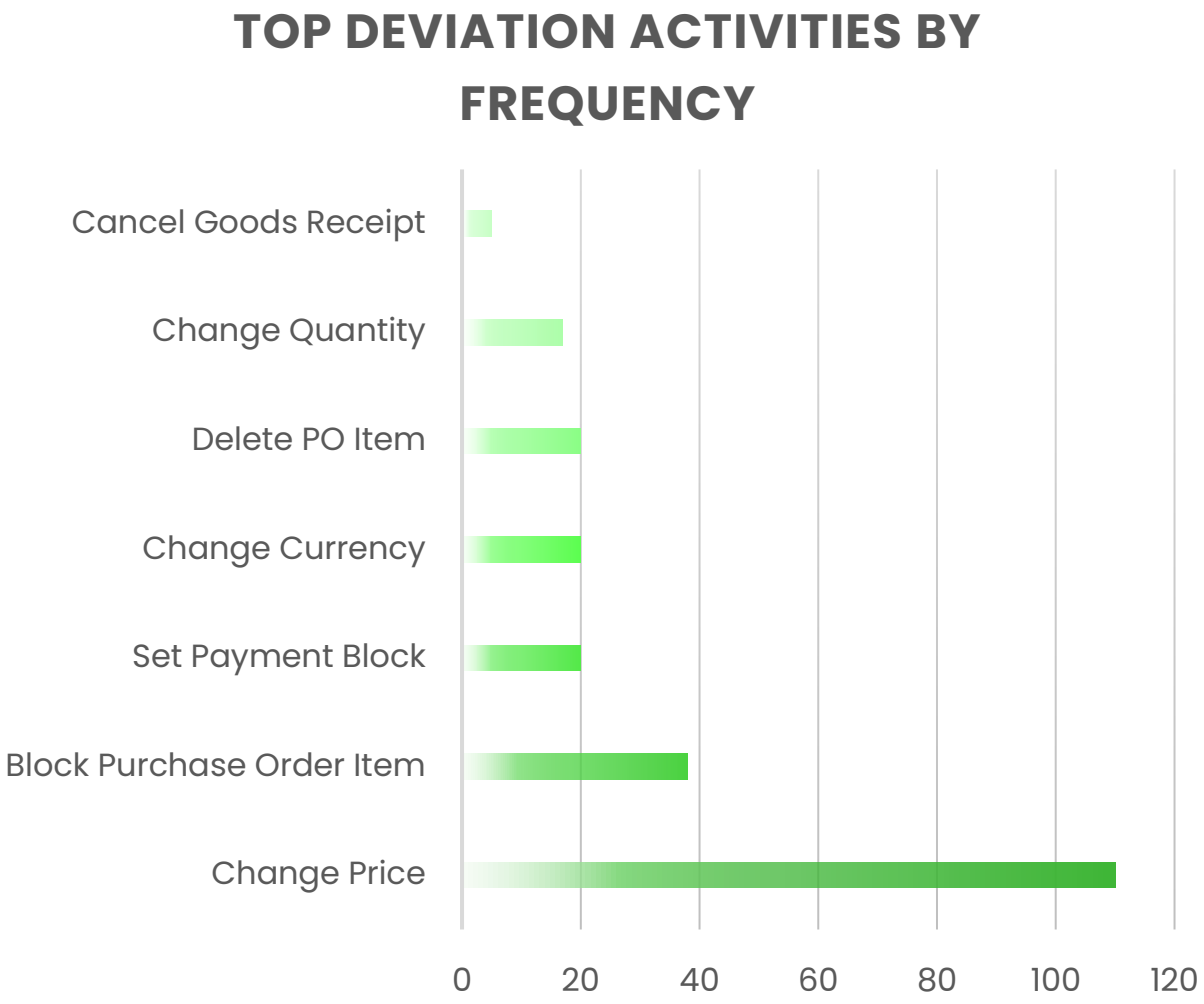
Process Composition

- Happy Path Cases: 537 (66.1%)
- Deviant Cases: 275 (33.9%)
- Deviation Events: 310 of 4,927 (6.3%)

Automation Profile

- Automated Events: 2,341 (47.5%)
- Manual Events: 2,586 (52.5%)
- Automation Gap: ▼ Target: 70%+

Where Does the Process Break Down?



⚠️ CRITICAL
110 price corrections
= highest rework type
(35.5% of all deviations)

🔒 BLOCKED
58 block events
(PO + Payment)
delay invoice clearance

📈 SCALE
33.9% of all cases
affected by at least
one deviation event

💡 OPPORTUNITY
Reducing 50% of
deviation events saves
~35 FTE-days annually

Opportunity 1: Resolve Payment Blocks & PO Holds

Challenge

- 20 Set Payment Block events across 20 unique PO cases
- 38 Block Purchase Order Item events = 58 total blocking events
- Each block requires manual investigation and resolution
- Blocks cause: delayed payments → missed early-pay discounts → strained vendor relationships → increased DPO

Solution

- Implement automated block detection via Celonis
- Parked & Blocked Invoices Starter Kit App
- Set up Action Flows: notify AP team + buyer when a block event occurs
- Automate goods receipt confirmation for trusted vendors
- Vendor alignment session for top-5 deviation vendors

Root Cause

- Price/quantity mismatches between PO and vendor invoice (upstream cause: Change Price events)
- Missing goods receipt confirmation auto-triggers block
- 52.5% manual activity rate = high human error exposure
- No automated block detection or escalation workflow
- Vendor 0000001001 drives 31% of Change Price events (upstream root cause of many blocks)

Assumptions:

$90K FTE cost/yr
15 min/block
Source: Celonis Value Framing Guide

Value

Blocking events		Effort per rework activity		FTE cost per minute		Improvement w/ Celonis		Committed Value
58	X	15 min	X	$0.42 / min	X	80%	=	**$292** Potential Savings p.a.

Opportunity 2: Eliminate Price Deviation Rework

Challenge

- 110 Change Price events = the #1 deviation in the entire process
- 13.5% of all PO items require a post-issuance price correction
- Vendor 0000001001: 34 events (31% of all Change Price)
- Vendor R1000: 21 events (19% of all Change Price)
- Top 2 vendors account for 50% of all price correction effort

Solution

- Master data fix: update vendor price lists in SAP for top-5 deviation vendors
- Implement 3-way match automation (PO–GR–Invoice)to catch mismatches before manual intervention needed
- Set up Celonis alerts for price delta > 5% threshold
- Quarterly vendor price reconciliation process
- Target: reduce Change Price events by 70%

Root Cause

ATTRIBUTE-BASED: Vendor-specific

→ Vendors 0000001001 & R1000 = 50% of price events

→ Price lists not synchronized between SAP and vendor ERP

PROCESS-BASED: No pre-PO price validation step

→ PO issued with catalog price

→ Vendor invoices at contracted/current price

→ Mismatch discovered only at invoice receipt stage

TIME-BASED: Corrections spike at period-end

→ Budget pressure drives faster (less accurate) PO creation

Assumptions:

$90K FTE cost/yr | 10 min/correction 70% realization (master data fix)

Value

Price correction events		Effort per rework activity		FTE cost per minute		Improvement w/ Celonis		Committed Value
110	X	10 min	X	$0.42 / min	X	70%	=	**$323** Potential Savings p.a.

Business Case Summary: AP Value Journey

Both opportunities target the Productivity pillar → impact P&L → improve Free Cash Flow

Objective	Opportunity	Owner	Time-To-Value	Framed	Committed	Realized
Productivity	Payment & PO Blocks	AP Team Lead	Short (1–2 sprints)	$292*	TBD	-
	Price Deviation Rework	AP Team Lead	Short (1–2 sprints)	$323*	TBD	-
Total Annual Impact (P&L)				**$615**	**-**	**-**
Total Spend Analyzed				**€207,096**		
Deviation Rate				**33.9% of cases**		
Current Automation				**47.5% (target: 70%+)**		

📌 METHODOLOGY NOTE

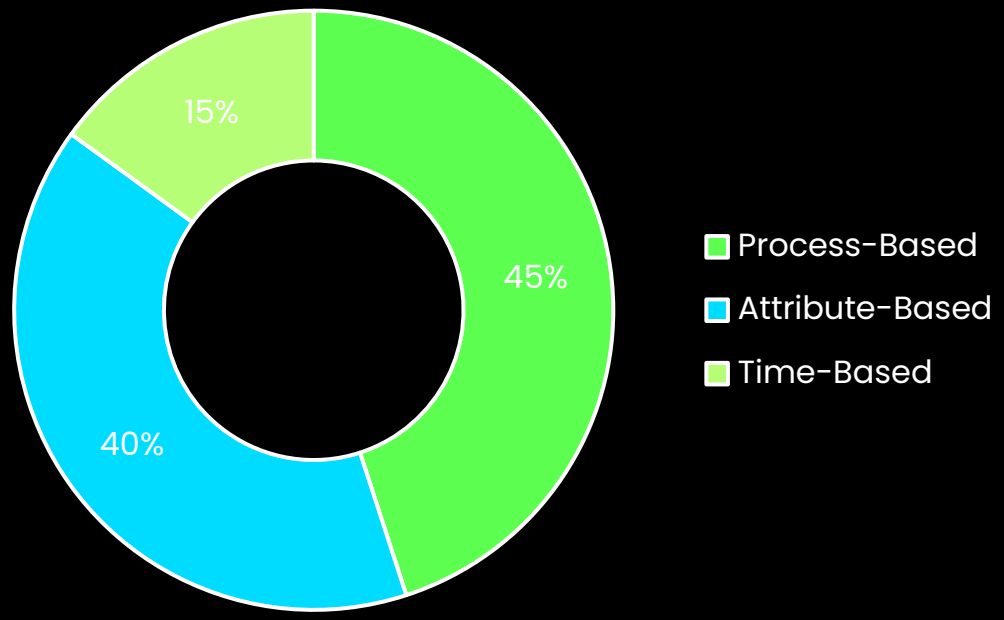
Value calculated using: Affected Events × Realization Potential × (Effort × FTE Cost/min)

FTE assumption: $90,000/year | $0.42/min | Source: Celonis Value Framing Guide

Root Cause Analysis: Who & Why?

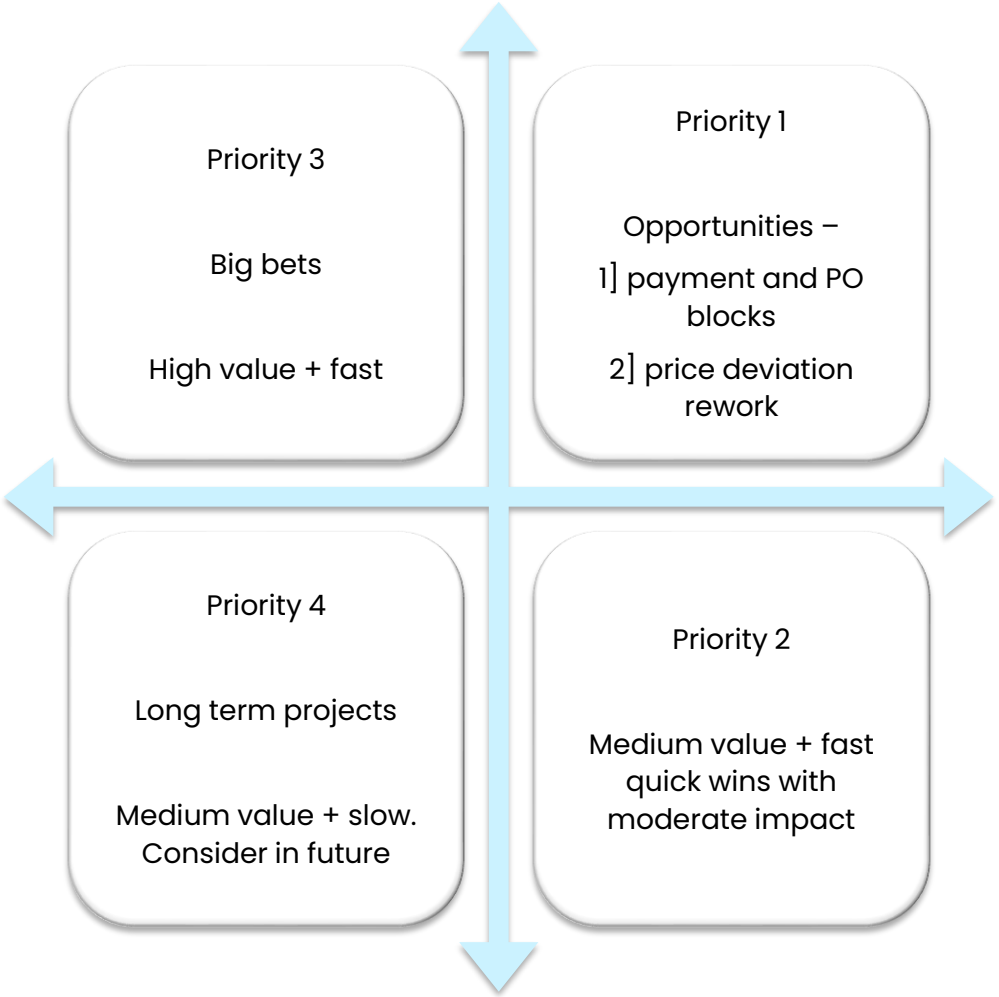
Vendor ID	Change price	Total	Risk level
0000001001	34	34	HIGH
R1000	21	21	HIGH
0000003021	12	12	MEDIUM
0000001011	10	10	MEDIUM
R3000	8	8	LOW

Root Cause Category



"Not all root causes are visible in the data.
Talking to business users reveals what the data can't show."
— Celonis Value Framing Methodology

Prioritization: Where to Act First



RECOMMENDED EXECUTION ORDER

PHASE 1 (Immediate — 0–4 weeks)

[1] Price Deviations

- Vendor master data fix (top 2 vendors)
- No technology required
- Eliminate 50% of problem immediately

PHASE 2 (Short-term — 4–12 weeks)

[2] Payment & PO Blocks

- Celonis Parked & Blocked Invoices App
- Automated detection + Action Flows
- Integrate with AP workflow

- Combined Annual Value: $615 (sample) | $2M–$2.7M (enterprise scale)
- Timeline to first value realization: 4–6 weeks

Thank You!